

Project Report: CSV-to-Warehouse Batch Pipeline - Data Engineer

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Family:	Data Engineer
Level:	Beginner
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Python ETL script: extract CSV, transform with pandas, load to PostgreSQL with logging.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Beginner]

Chapter focus: Project Overview** *(Level: Beginner)

CSV-to-Warehouse Batch Pipeline** is a beginner-level end-to-end project for Data Engineer. **Duration:** 2 weeks. Python ETL script: extract CSV, transform with pandas, load to PostgreSQL with logging. **Tools:** Python, pandas, PostgreSQL.

Code Reference:

```
Objective: Extract from CSV/API
Objective: Transform and validate
Objective: Load to PostgreSQL
Objective: Idempotent daily run
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Data Engineer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Data Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Extract [Beginner]

Chapter focus: Phase: Extract** *(Level: Beginner)

Read CSV with pandas. Complete these tasks: Schema detection; Error rows. Deliverable for this phase: Raw landing table.

Code Reference:

- [] Schema detection
- [] Error rows

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Extract is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Data Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Raw landing table to the teacher.

Chapter 3: Phase: Transform [Beginner]

Chapter focus: Phase: Transform** *(Level: Beginner)

Clean and enrich. Complete these tasks: Dedup; Type cast. Deliverable for this phase: Staging table.

Code Reference:

- [] Dedup
- [] Type cast

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Transform is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Data Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Staging table to the teacher.

Chapter 4: Phase: Load [Beginner]**Chapter focus: Phase: Load** *(Level: Beginner)**

Upsert to warehouse. Complete these tasks: ON CONFLICT; Transaction. Deliverable for this phase: Final table populated.

Code Reference:

- [] ON CONFLICT
- [] Transaction

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Load is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Data Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Final table populated to the teacher.

Chapter 5: Phase: Schedule [Beginner]**Chapter focus: Phase: Schedule** *(Level: Beginner)**

Cron + logging. Complete these tasks: Log rotation; Alert on fail. Deliverable for this phase: Scheduled job running.

Code Reference:

- [] Log rotation
- [] Alert on fail

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Schedule is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Data Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Scheduled job running to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Beginner]

Chapter focus: Final Deliverables and Review *(Level: Beginner)**

Submit all final deliverables: Pipeline script; SQL DDL; Run log sample. Skills demonstrated: ETL, Python, SQL. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

Deliverable: Pipeline script

Deliverable: SQL DDL

Deliverable: Run log sample

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide**What it actually is**

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Pipeline script and passes the rubric.